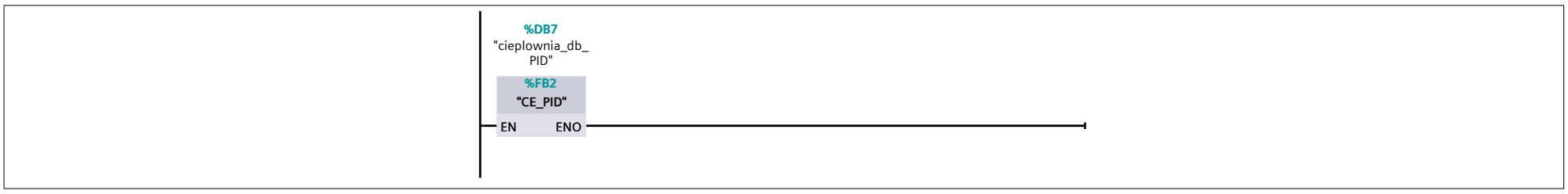
100ms [OB30]

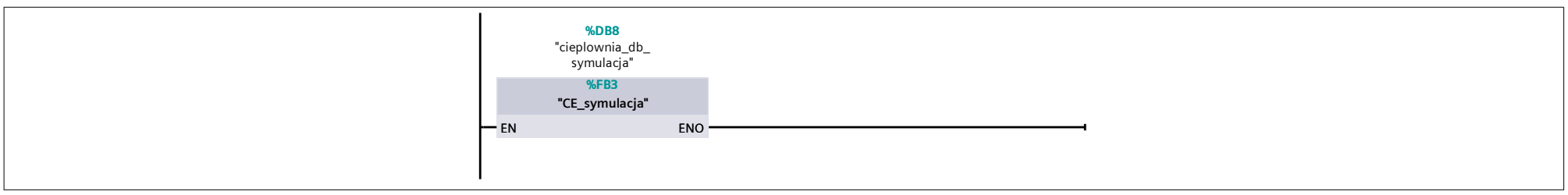
100ms Properties							
General							
Name	100ms	**Number**	30	**Type**	OB	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Initial_Call	Bool		Initial call of this OB
Event_Count	Int		Events discarded
Temp			
Constant			

Network 1: PID



Network 2: Symulacja



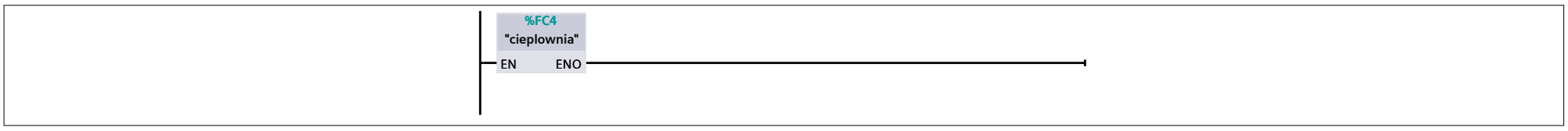
Main [OB1]

Main Properties							
General							
Name	Main	**Number**	1	**Type**	OB	**Language**	LAD
Numbering	Automatic						
Information							
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

Network 1:

wywolanie glownego programu cieplowni

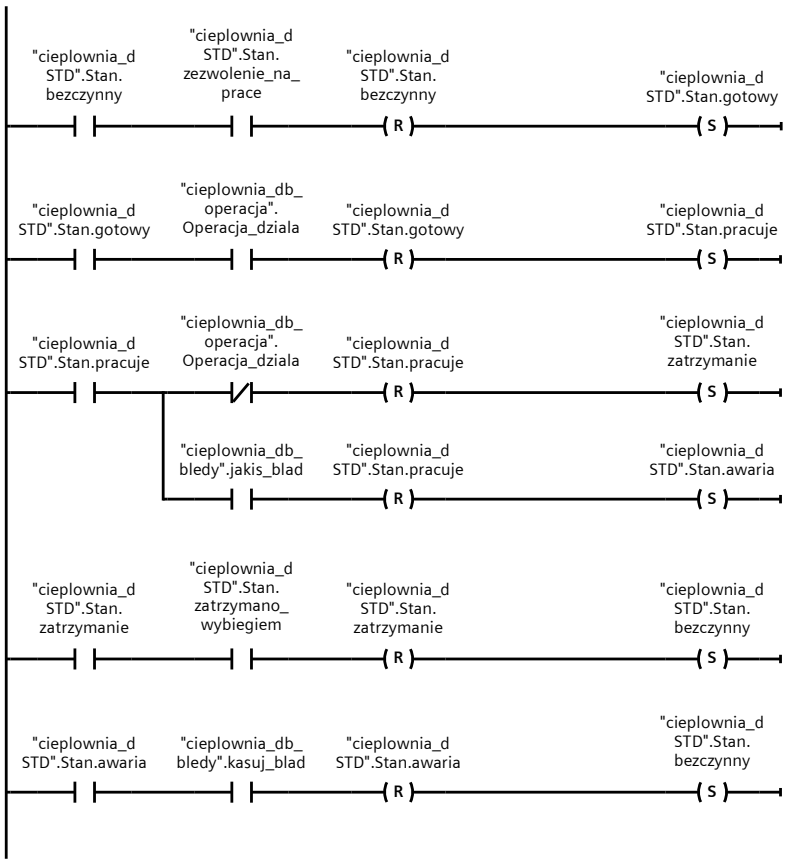


cieplownia [FC4]

cieplownia Properties							
General							
Name	cieplownia	**Number**	4	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

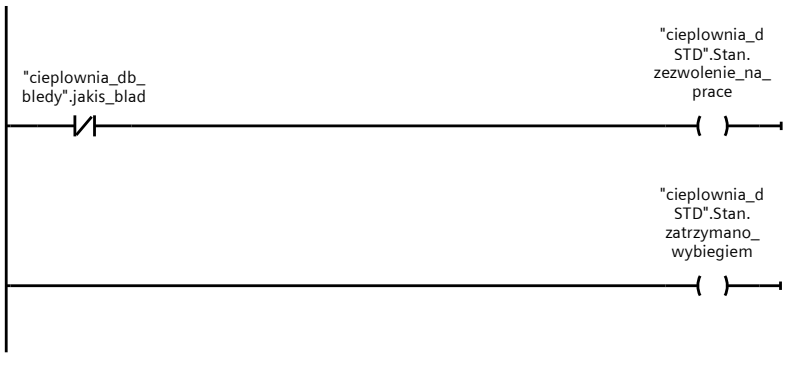
Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
▼ Return			
cieplownia	Void		

Network 1: stan urzadzenia



Network 2: stan urzadzenia v2

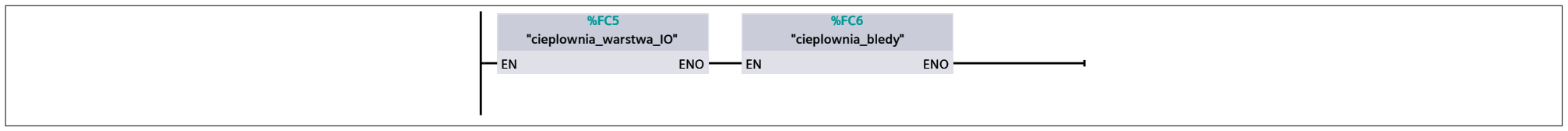
warunki na rozpoczęcie i zatrzymanie pracy



Network 3: wywolanie komponentow

```
0001 "zawor_bistabilny"("cieplownia_db_komponenty".zawor_V1);
0002
0003 "czujnik"(bit_symulacji:= "cieplownia_dSTD".Symulacja,
0004           czujnik:="cieplownia_db_komponenty".czujnik_P1);
0005
0006 "czujnik"(bit_symulacji:= "cieplownia_dSTD".Symulacja,
0007           czujnik:="cieplownia_db_komponenty".czujnik_T1);
0008
0009 "zawor_analogowy+pompa"("cieplownia_db_komponenty".zawor_an_V2);
0010
0011 "zawor_analogowy+pompa"("cieplownia_db_komponenty".pompa_P1);
```

Network 4: wywolanie warstwy IO i bledow



Network 5: wywolania blokow mieszkalnych

```
0001 "cieplownia_db_dane".zadania_blokow.blok_1 := "Clock_1Hz";
0002 "cieplownia_db_dane".zadania_blokow.blok_2 := "Clock_0.625Hz";
0003 "cieplownia_db_dane".zadania_blokow.blok_3 :=
0004 "cieplownia_db_dane".zadania_blokow.blok_4 := "Clock_0.5Hz";
```

Network 6: wywolanie operacji

cieplownia_bledy [FC6]

cieplownia_bledy Properties							
General							
Name	cieplownia_bledy	**Number**	6	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

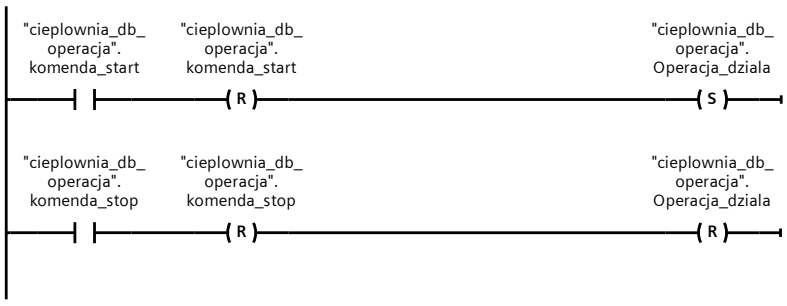
Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
▼ Return			
cieplownia_bledy	Void		

cieplownia_operacja [FC7]

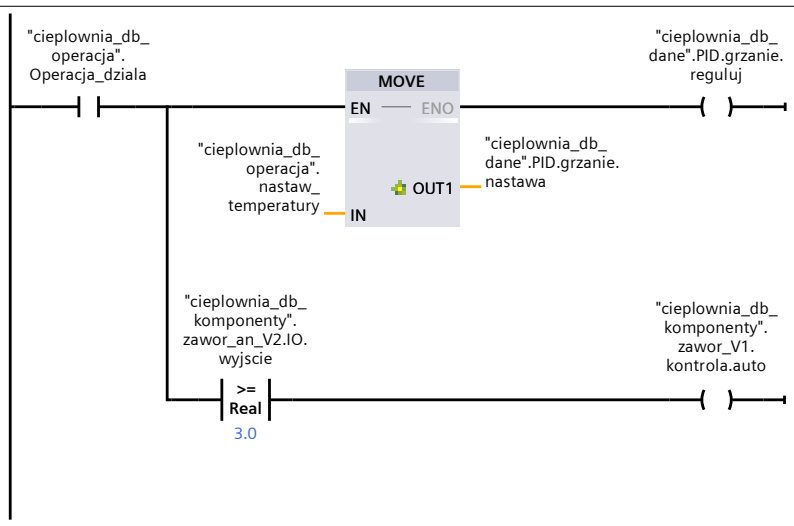
cieplownia_operacja Properties							
General							
Name	cieplownia_operacja	**Number**	7	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
▼ Return			
cieplownia_operacja	Void		

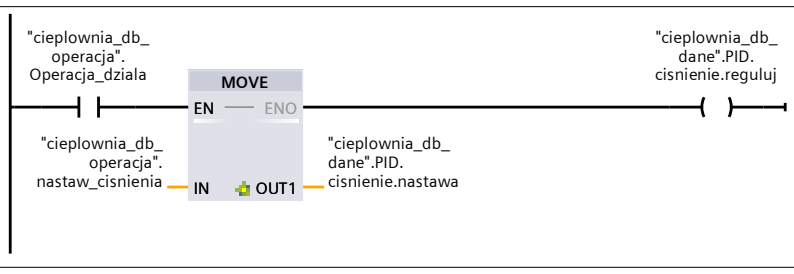
Network 1: Start / Stop operacji



Network 2: Grzanie parą



Network 3: Utrzymywanie ciśnienia



cieplownia_warstwa_IO [FC5]

cieplownia_warstwa_IO Properties							
General							
Name	cieplownia_warstwa_IO	**Number**	5	**Type**	FC	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
▼ Return			
cieplownia_warstwa_IO	Void		

CE_PID [FB2]

CE_PID Properties							
General							
Name	CE_PID	**Number**	2	**Type**	FB	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

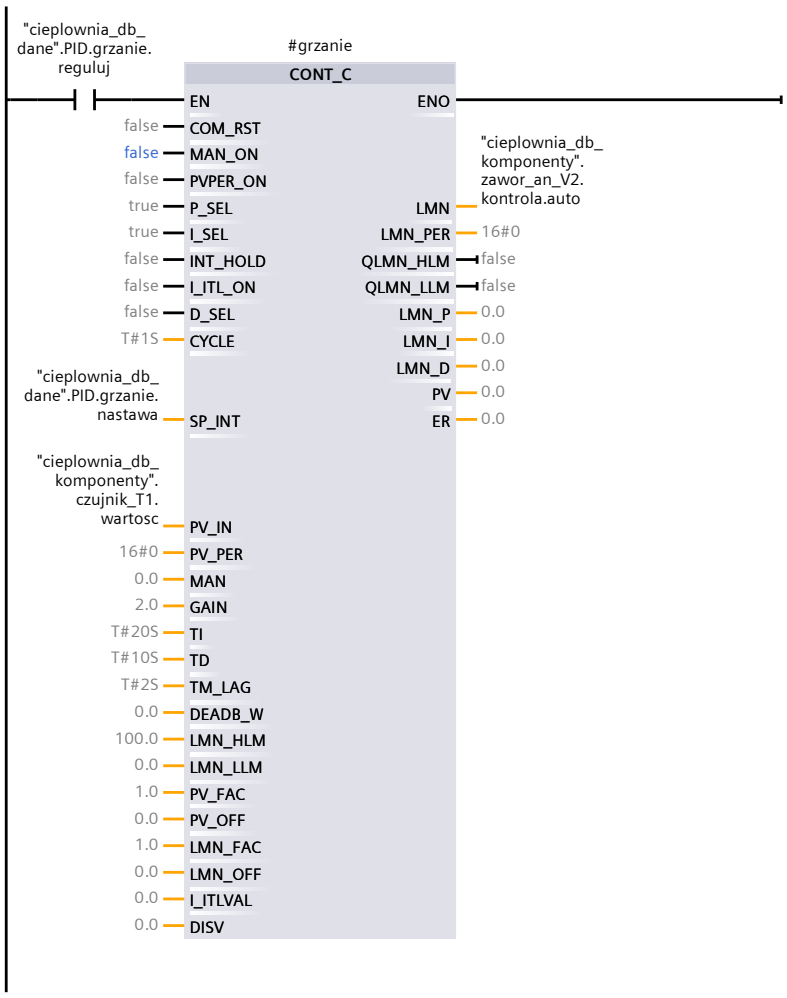
Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
Output									
InOut									
▼ Static									
▼ grzanie	CONT_C			False	False	False	False		
▼ Input									
COM_RST	Bool	false	Retain	False	False	False	False		complete restart
MAN_ON	Bool	true	Retain	False	False	False	False		manual value on
PVPER_ON	Bool	false	Retain	False	False	False	False		process variable peripherie on
P_SEL	Bool	true	Retain	False	False	False	False		proportional action on
I_SEL	Bool	true	Retain	False	False	False	False		integral action on
INT_HOLD	Bool	false	Retain	False	False	False	False		integral action hold
I_ITL_ON	Bool	false	Retain	False	False	False	False		initialization of the integral action
D_SEL	Bool	false	Retain	False	False	False	False		derivative action on
CYCLE	Time	T#1S	Retain	False	False	False	False		sample time
SP_INT	Real	0.0	Retain	False	False	False	False		internal setpoint
PV_IN	Real	0.0	Retain	False	False	False	False		process variable in
PV_PER	Word	16#0	Retain	False	False	False	False		process variable peripherie
MAN	Real	0.0	Retain	False	False	False	False		manual value
GAIN	Real	2.0	Retain	False	False	False	False		proportional gain
TI	Time	T#20S	Retain	False	False	False	False		reset time
TD	Time	T#10S	Retain	False	False	False	False		derivative time
TM_LAG	Time	T#2S	Retain	False	False	False	False		time lag of the derivative action
DEADB_W	Real	0.0	Retain	False	False	False	False		dead band width
LMN_HLM	Real	100.0	Retain	False	False	False	False		manipulated value high limit
LMN_LLM	Real	0.0	Retain	False	False	False	False		manipulated value low limit
PV_FAC	Real	1.0	Retain	False	False	False	False		process variable factor
PV_OFF	Real	0.0	Retain	False	False	False	False		process variable offset
LMN_FAC	Real	1.0	Retain	False	False	False	False		manipulated value factor
LMN_OFF	Real	0.0	Retain	False	False	False	False		manipulated value offset
I_ITLVAL	Real	0.0	Retain	False	False	False	False		initialization value of the integral action
DISV	Real	0.0	Retain	False	False	False	False		disturbance variable
▼ Output									
LMN	Real	0.0	Retain	False	False	False	False		manipulated value
LMN_PER	Word	16#0	Retain	False	False	False	False		manipulated value peripherie
QLMN_HLM	Bool	false	Retain	False	False	False	False		high limit of manipulated value reached
QLMN_LLM	Bool	false	Retain	False	False	False	False		low limit of manipulated value reached
LMN_P	Real	0.0	Retain	False	False	False	False		proportionality component
LMN_I	Real	0.0	Retain	False	False	False	False		integral component
LMN_D	Real	0.0	Retain	False	False	False	False		derivative component
PV	Real	0.0	Retain	False	False	False	False		process variable
ER	Real	0.0	Retain	False	False	False	False		error signal
InOut									
▼ Static									
sInvAlt	Real	0.0	Retain	False	False	False	False		
sIanteilAlt	Real	0.0	Retain	False	False	False	False		
sRestInt	Real	0.0	Retain	False	False	False	False		
sRestDif	Real	0.0	Retain	False	False	False	False		
sRueck	Real	0.0	Retain	False	False	False	False		
sLmn	Real	0.0	Retain	False	False	False	False		
sbArwHLmOn	Bool	false	Retain	False	False	False	False		
sbArwLLmOn	Bool	false	Retain	False	False	False	False		
sbILimOn	Bool	true	Retain	False	False	False	False		
▼ cisnienie	CONT_C			False	False	False	False		
▼ Input									

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
COM_RST	Bool	false	Retain	False	False	False	False		complete restart
MAN_ON	Bool	true	Retain	False	False	False	False		manual value on
PVPER_ON	Bool	false	Retain	False	False	False	False		process variable peripherie on
P_SEL	Bool	true	Retain	False	False	False	False		proportional action on
I_SEL	Bool	true	Retain	False	False	False	False		integral action on
INT_HOLD	Bool	false	Retain	False	False	False	False		integral action hold
I_ITL_ON	Bool	false	Retain	False	False	False	False		initialization of the integral action
D_SEL	Bool	false	Retain	False	False	False	False		derivative action on
CYCLE	Time	T#1S	Retain	False	False	False	False		sample time
SP_INT	Real	0.0	Retain	False	False	False	False		internal setpoint
PV_IN	Real	0.0	Retain	False	False	False	False		process variable in
PV_PER	Word	16#0	Retain	False	False	False	False		process variable peripherie
MAN	Real	0.0	Retain	False	False	False	False		manual value
GAIN	Real	2.0	Retain	False	False	False	False		proportional gain
TI	Time	T#20S	Retain	False	False	False	False		reset time
TD	Time	T#10S	Retain	False	False	False	False		derivative time
TM_LAG	Time	T#2S	Retain	False	False	False	False		time lag of the derivative action
DEADB_W	Real	0.0	Retain	False	False	False	False		dead band width
LMN_HLM	Real	100.0	Retain	False	False	False	False		manipulated value high limit
LMN_LLM	Real	0.0	Retain	False	False	False	False		manipulated value low limit
PV_FAC	Real	1.0	Retain	False	False	False	False		process variable factor
PV_OFF	Real	0.0	Retain	False	False	False	False		process variable offset
LMN_FAC	Real	1.0	Retain	False	False	False	False		manipulated value factor
LMN_OFF	Real	0.0	Retain	False	False	False	False		manipulated value offset
I_ITLVAL	Real	0.0	Retain	False	False	False	False		initialization value of the integral action
DISV	Real	0.0	Retain	False	False	False	False		disturbance variable
▼ Output									
LMN	Real	0.0	Retain	False	False	False	False		manipulated value
LMN_PER	Word	16#0	Retain	False	False	False	False		manipulated value peripherie
QLMN_HLM	Bool	false	Retain	False	False	False	False		high limit of manipulated value reached
QLMN_LLM	Bool	false	Retain	False	False	False	False		low limit of manipulated value reached
LMN_P	Real	0.0	Retain	False	False	False	False		proportionality component
LMN_I	Real	0.0	Retain	False	False	False	False		integral component
LMN_D	Real	0.0	Retain	False	False	False	False		derivative component
PV	Real	0.0	Retain	False	False	False	False		process variable
ER	Real	0.0	Retain	False	False	False	False		error signal
InOut									
▼ Static									
sInvAlt	Real	0.0	Retain	False	False	False	False		
sIanteilAlt	Real	0.0	Retain	False	False	False	False		
sRestInt	Real	0.0	Retain	False	False	False	False		
sRestDif	Real	0.0	Retain	False	False	False	False		
sRueck	Real	0.0	Retain	False	False	False	False		
sLmn	Real	0.0	Retain	False	False	False	False		
sbArwHLmOn	Bool	false	Retain	False	False	False	False		
sbArwLLmOn	Bool	false	Retain	False	False	False	False		
sbILimOn	Bool	true	Retain	False	False	False	False		
Temp									
Constant									

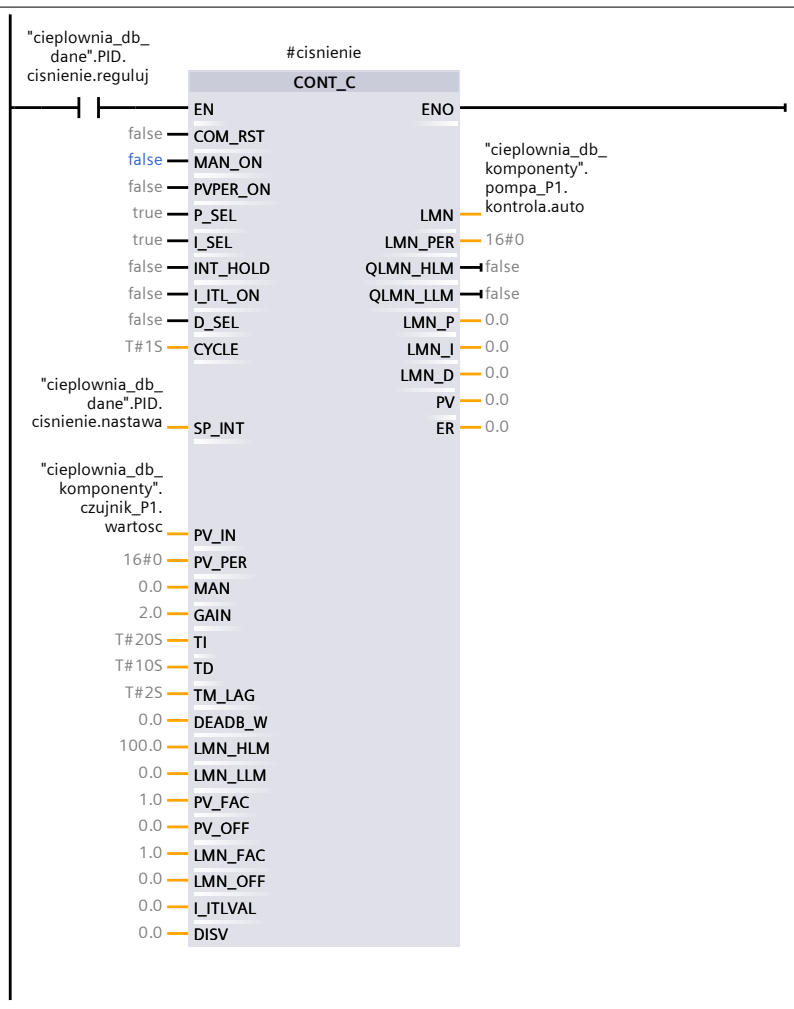
Network 1:

```
0001 "cieplownia_db_komponenty".zawor_an_V2.kontrola.auto := 0;
0002 "cieplownia_db_komponenty".pompa_P1.kontrola.auto := 0;
```

Network 2: Regulacja temperatury (grzania)



Network 3: Regulacja ciśnienia



CE_symulacja [FB3]

CE_symulacja Properties							
General							
Name	CE_symulacja	**Number**	3	**Type**	FB	**Language**	LAD
Numbering	Automatic						
Information							
Title		**Author**		**Comment**		**Family**	
Version	0.1	**User-defined ID**					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Setpoint	Supervision	Comment
Input									
Output									
InOut									
Static									
temperatura	Real	0.0	Non-retain	True	True	True	False		
cisnienie	Real	0.0	Non-retain	True	True	True	False		
Temp									
Constant									

Network 1: TT1 czujnik temperatury

```
0001 //odbiór ciepła przez bloki
0002 IF "cieplownia_db_dane".zadania_blokow.blok_1 OR "cieplownia_db_dane".zadania_blokow.blok_2
0003     OR "cieplownia_db_dane".zadania_blokow.blok_3 OR "cieplownia_db_dane".zadania_blokow.blok_4
0004 THEN
0005     #temperatura := #temperatura - 0.01;
0006 END_IF;
0007
0008
0009 //grzanie pary przez zawór
0010 IF "cieplownia_db_komponenty".zawor_V1.IO.wyjscie
0011 THEN
0012     #temperatura := #temperatura + ("cieplownia_db_komponenty".zawor_an_V2.IO.wyjscie / 1000);
0013 END_IF;
0014
0015 //ograniczenia
0016 IF #temperatura < 0.0
0017 THEN
0018     #temperatura := 0;
0019 END_IF;
0020
0021 IF #temperatura > 100.0
0022 THEN
0023     #temperatura := 100.0;
0024 END_IF;
0025
0026 //przepisać wartość do komponentu
0027
0028 "cieplownia_db_komponenty".czujnik_T1.IO.wejscie_symulowane := #temperatura;
```

Network 2: PT1 czujnik ciśnienia

```
0001 //spadek ciśnienia przez odbiór
0002 IF "cieplownia_db_dane".zadania_blokow.blok_1 OR "cieplownia_db_dane".zadania_blokow.blok_2
0003     OR "cieplownia_db_dane".zadania_blokow.blok_3 OR "cieplownia_db_dane".zadania_blokow.blok_4
0004 THEN
0005     #cisnienie := #cisnienie - 0.025;
0006 END_IF;
0007
0008
0009 //dobijanie cisnienia przez pompe
0010 IF "cieplownia_db_komponenty".pompa_P1.kontrola.auto > 0.0
0011 THEN
0012     #cisnienie := #cisnienie + ("cieplownia_db_komponenty".pompa_P1.IO.wyjscie / 1000);
0013 END_IF;
0014
0015 //ograniczenia
0016 IF #cisnienie < 0.0
0017 THEN
0018     #cisnienie := 0;
0019 END_IF;
0020
0021 IF #cisnienie > 10.0
0022 THEN
0023     #cisnienie := 10.0;
```

```
0024 END_IF;
0025
0026 //przepisać wartość do komponentu
0027
0028 "cieplownia_db_komponenty".czujnik_P1.IO.wejscie_symulowane := #cisnienie;
```